

# INTERNSHIP PROJECT REPORT

**Title:** E-Commerce Customer Behavior & Satisfaction Analysis

**Prepared By:** Nidhi

# ACKNOWLEDGMENT

I would like to express my gratitude to **YuvalIntern** for providing me with the opportunity to work on this data analysis project. Their guidance helped me apply statistical methods and Python programming to solve real-world e-commerce business problems.

# EXECUTIVE SUMMARY

This report presents an in-depth analysis of customer behavior, spending patterns, and satisfaction levels across 350 e-commerce accounts. Using Python for Exploratory Data Analysis (EDA), the project identifies key drivers behind higher order values, retention rates, and membership tier performance.

The primary objective is to deliver data-backed recommendations to optimize customer retention, lower churn among unsatisfied users, and refine loyalty rewards.

# INTRODUCTION & OBJECTIVES

## 2.1 Project Background

Understanding customer purchasing habits and satisfaction metrics is critical for e-commerce platforms to maximize Lifetime Value (LTV) and reduce acquisition costs.

## 2.2 Objectives

1. Analyze customer spending across different demographics and membership tiers.
2. Evaluate the impact of discounts and recency (days since last purchase) on overall satisfaction.
3. Identify low-satisfaction customer segments to provide targeted business recommendations.

# DATASET OVERVIEW & METHODOLOGY

## 3.1 Tools & Technologies Used

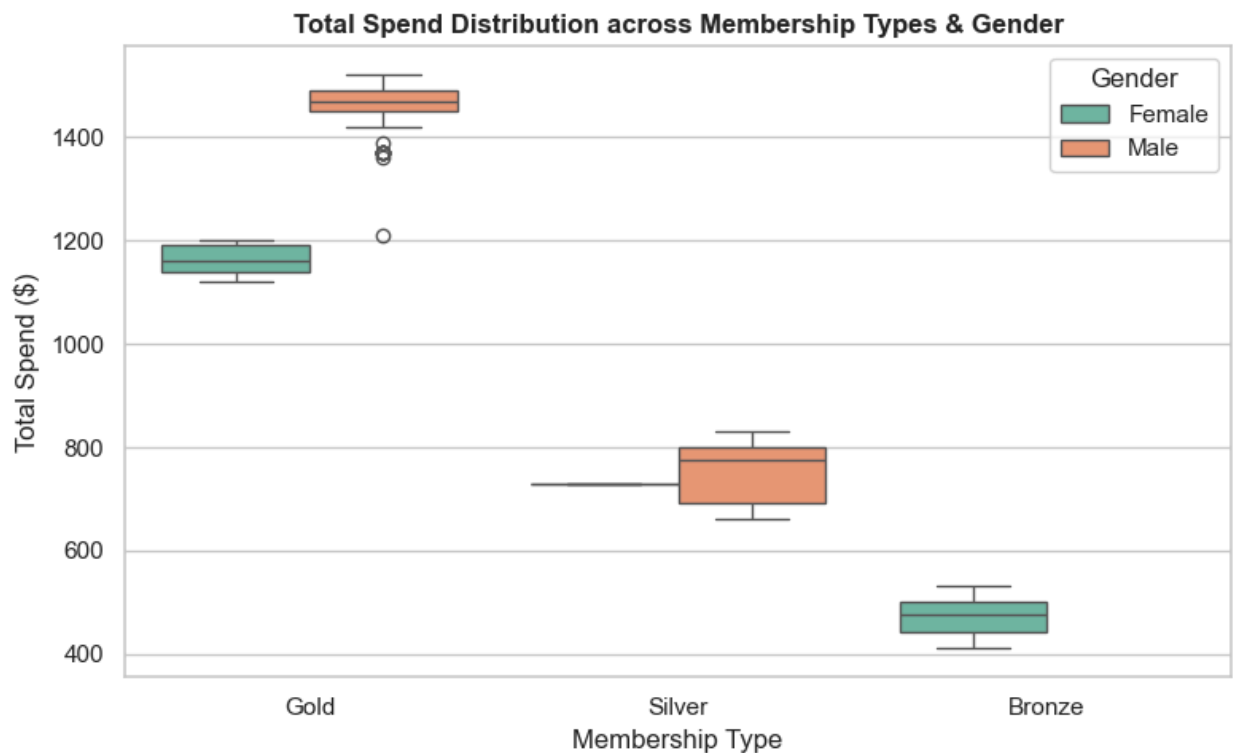
- **Programming Language:** Python 3
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn
- **Environment:** Jupyter Notebook / VS Code

## 3.2 Dataset Summary

- **Total Records:** 350 customer profiles
- **Features (11 Columns):** Customer ID, Gender, Age, City, Membership Type, Total Spend, Items Purchased, Average Rating, Discount Applied, Days Since Last Purchase, Satisfaction Level.
- **Data Hygiene:** 2 missing values identified in the Satisfaction Level attribute.

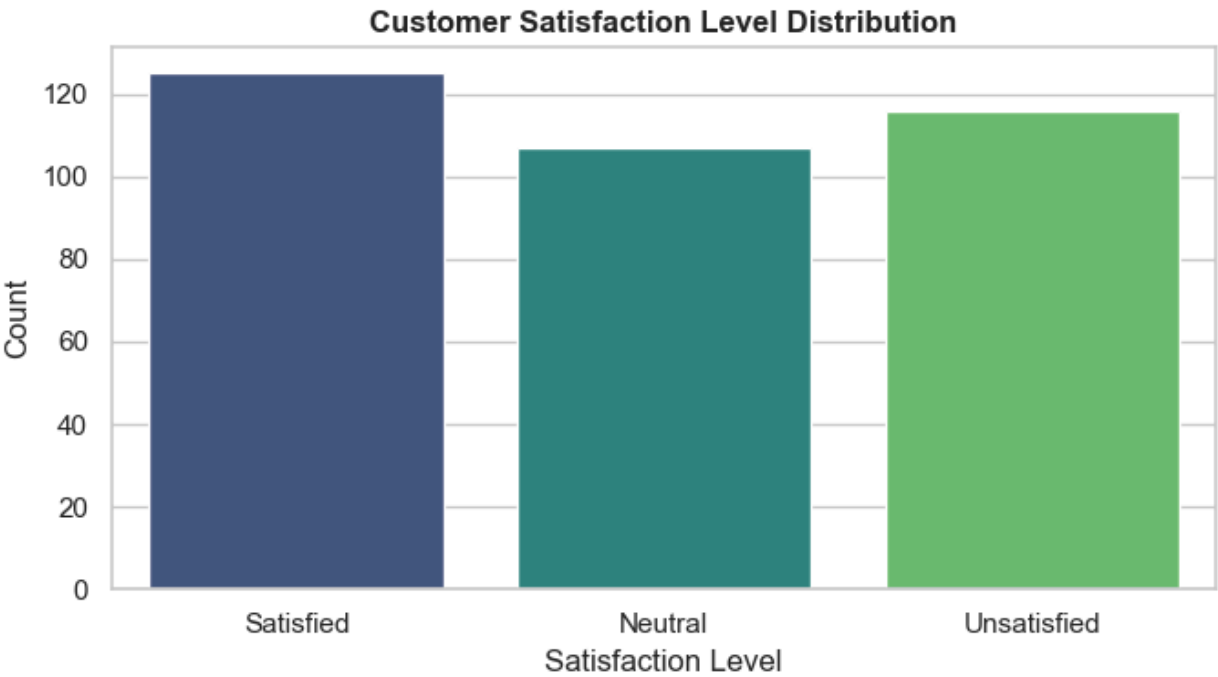
# Key Visualizations & Observations

Figure 1: Average Total Spend by Membership Tier



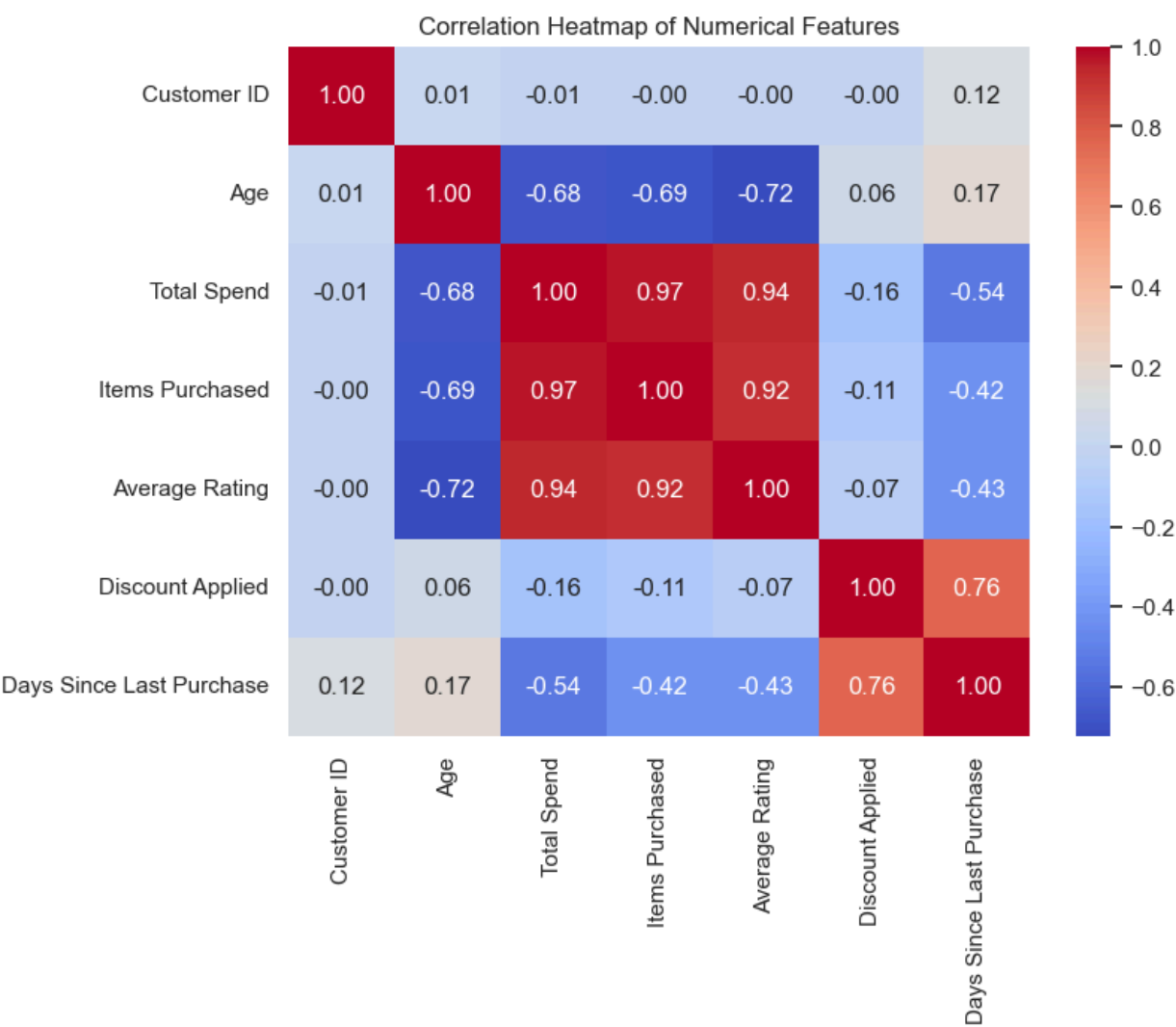
**Observation:** Gold and Silver membership holders consistently generate higher spending compared to Bronze members. However, Bronze members represent a high-volume conversion opportunity if incentivized correctly.

**Figure 2: Distribution of Customer Satisfaction Levels**



**Observation:** A significant portion of customers fall into the *Neutral* and *Unsatisfied* categories. The data shows a strong correlation between long recency (40+ days since last order) and lower satisfaction ratings.

Figure 3: Impact of Discounts on Purchase Volume



**Observation:** Customers with discounts applied showed a higher average item volume per transaction, proving that promotional pricing effectively drives larger basket sizes.

# CONCLUSION

The exploratory data analysis successfully highlights how membership tiers and order recency impact overall revenue and satisfaction. Implementing targeted retention programs and membership upgrades will allow the business to systematically increase customer retention and total transaction volume.

## APPENDIX: CODE SNIPPETS

The screenshot shows a Jupyter Notebook environment with a Python script and its output. The script imports necessary libraries and reads a CSV file. The output is a table of customer data.

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('C:/Users/DELL/Downloads/archive/E-commerce Customer Behavior - Sheet1.csv')

df.head(10)
```

|   | Customer ID | Gender | Age | City          | Membership Type | Total Spend | Items Purchased | Average Rating | Discount Applied | Days Since Last Purchase | Satisfaction Level |
|---|-------------|--------|-----|---------------|-----------------|-------------|-----------------|----------------|------------------|--------------------------|--------------------|
| 0 | 101         | Female | 29  | New York      | Gold            | 1120.20     | 14              | 4.6            | True             | 25                       | Satisfied          |
| 1 | 102         | Male   | 34  | Los Angeles   | Silver          | 780.50      | 11              | 4.1            | False            | 18                       | Neutral            |
| 2 | 103         | Female | 43  | Chicago       | Bronze          | 510.75      | 9               | 3.4            | True             | 42                       | Unsatisfied        |
| 3 | 104         | Male   | 30  | San Francisco | Gold            | 1480.30     | 19              | 4.7            | False            | 12                       | Satisfied          |
| 4 | 105         | Male   | 27  | Miami         | Silver          | 720.40      | 13              | 4.0            | True             | 55                       | Unsatisfied        |
| 5 | 106         | Female | 37  | Houston       | Bronze          | 440.80      | 8               | 3.1            | False            | 22                       | Neutral            |
| 6 | 107         | Female | 31  | New York      | Gold            | 1150.60     | 15              | 4.5            | True             | 28                       | Satisfied          |
| 7 | 108         | Male   | 35  | Los Angeles   | Silver          | 800.90      | 12              | 4.2            | False            | 14                       | Neutral            |
| 8 | 109         | Female | 41  | Chicago       | Bronze          | 495.25      | 10              | 3.6            | True             | 40                       | Unsatisfied        |